Spatio-Temporal Momentum — Backtest Report

Model: STM (TFT)
Period: 2026-06-01 to 2026-09-29 (84 sessions)
Universe (20): NWSA, FOX, HAS, AAP, BG, COP, PSX, MCO, CBOE, MCK, INCY, EMR, ETN, TDY, AMCR, FCX, ESS, WY, ED, CMS

Layout follows Tan, Roberts & Zohren (2023), arXiv:2302.10175.

Notes:
- DMN (reference model of the paper) is not implemented in this project and is not reported.
- TSMOM / CSMOM use the longest configured return horizon (21 days); the paper uses 252 days.
- Rescaled results target 15% annualised volatility with an ex-ante 20-day EWM estimate.
- Rolling correlations use a 21-session window (paper: 1 year).
- SHAP values: Expected Gradients (shap.GradientExplainer formulation); the paper uses Deep SHAP.

Table 1: Performance Metrics for Strategies – Raw Signal Outputs

	E[Return]	Vol.	Downside Deviation	MDD	Sharpe	Sortino	Calmar	Hit Rate	Ave. P / Ave. L
Long Only	0.003	0.051	0.035	0.041	0.058	0.084	0.071	0.536	0.875
TSMOM	-0.013	0.042	0.030	0.030	-0.309	-0.435	-0.435	0.548	0.785
MACD	0.002	0.020	0.014	0.008	0.082	0.116	0.213	0.512	0.966
CSMOM	-0.036	0.018	0.013	0.019	-2.062	-2.728	-1.940	0.393	1.121
STM (TFT)	-0.021	0.020	0.015	0.014	-1.034	-1.407	-1.476	0.536	0.736

Table 2: Performance Metrics for Strategies – Rescaled to Target Volatility

	E[Return]	Vol.	Downside Deviation	MDD	Sharpe	Sortino	Calmar	Hit Rate	Ave. P / Ave. L
Long Only	-0.074	0.158	0.116	0.141	-0.464	-0.635	-0.523	0.562	0.722
TSMOM	0.093	0.159	0.104	0.079	0.582	0.892	1.178	0.531	0.973
MACD	0.039	0.148	0.101	0.055	0.264	0.387	0.711	0.516	0.983
CSMOM	-0.198	0.159	0.112	0.116	-1.246	-1.770	-1.707	0.391	1.283
STM (TFT)	0.084	0.151	0.105	0.053	0.558	0.804	1.589	0.562	0.851

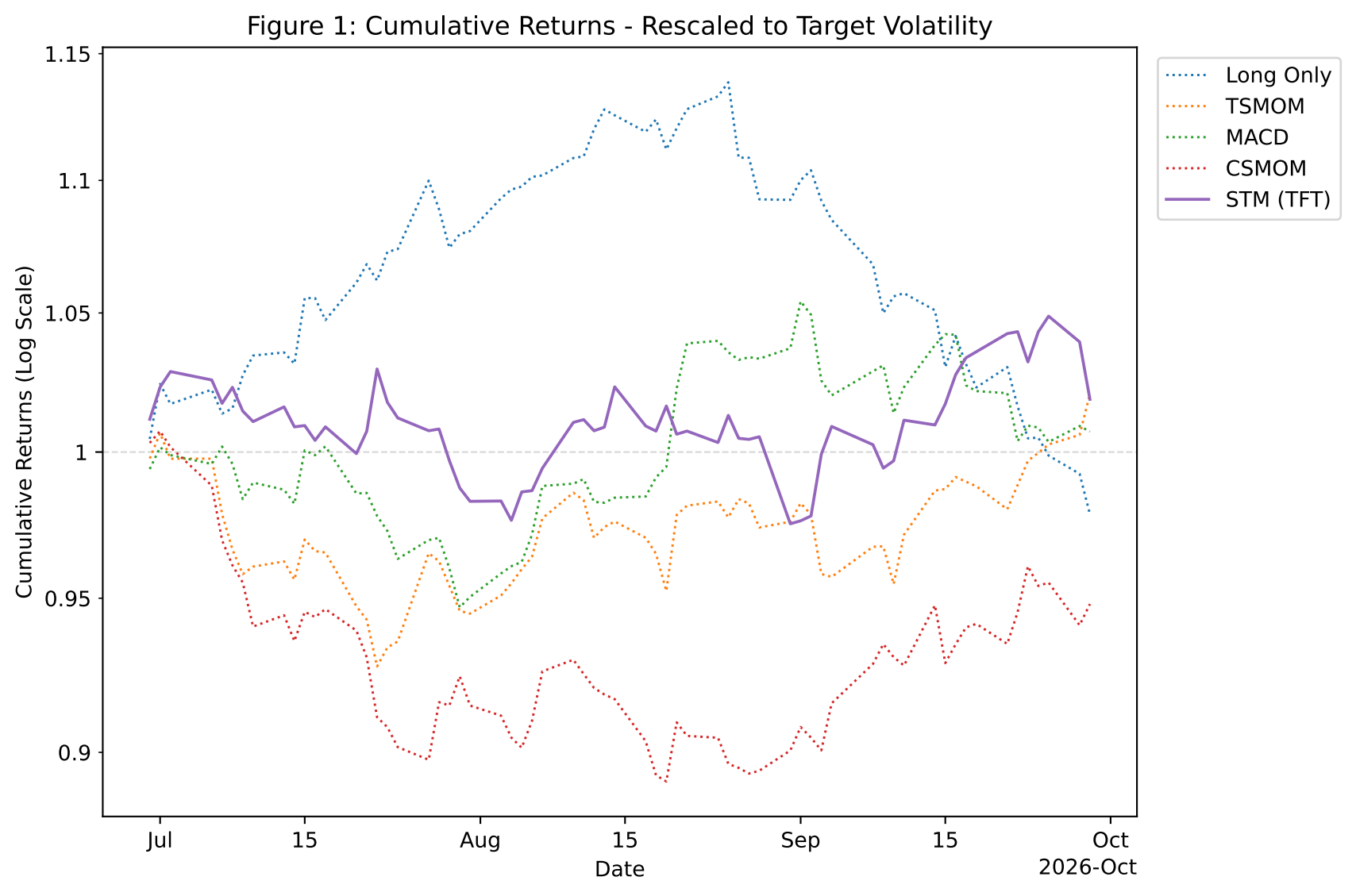
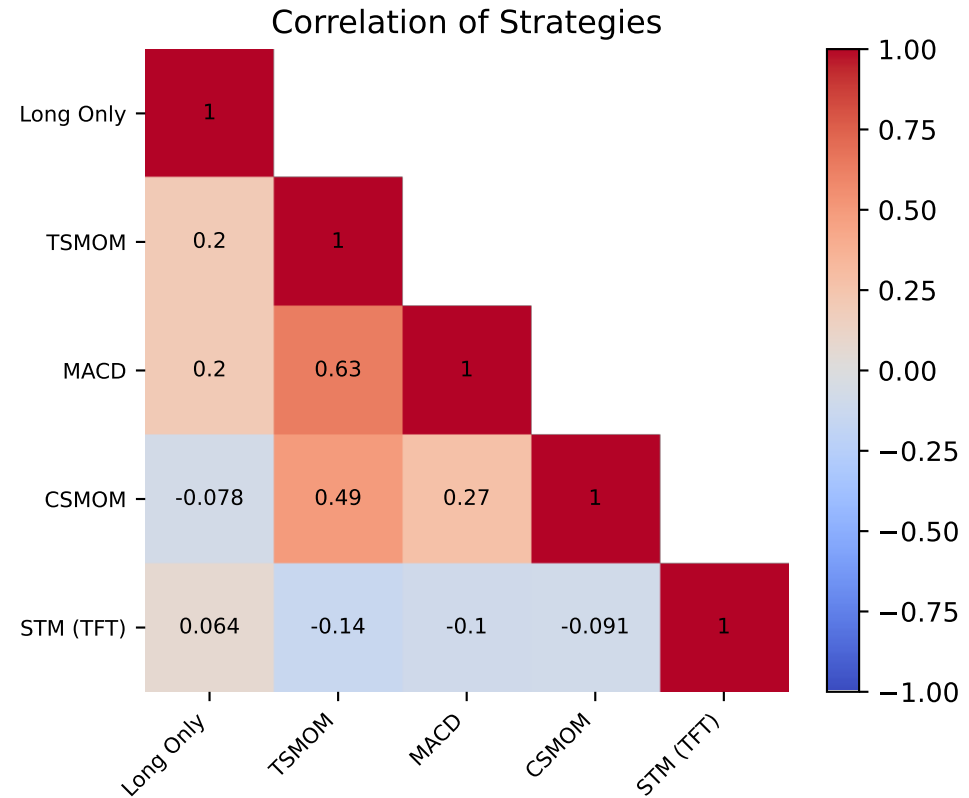

Figure 1: Cumulative Returns - Rescaled to Target Volatility
Cumulative Returns (Log Scale)
1.15
1.1
1.05
1
0.95
0.9
Jul
15
Aug
15
Sep
15
Oct
Date
2026-Oct
Long Only
TSMOM
MACD
CSMOM
STM (TFT)

Figure 3: (Left) Correlation of Strategies (Right) Rolling Correlations



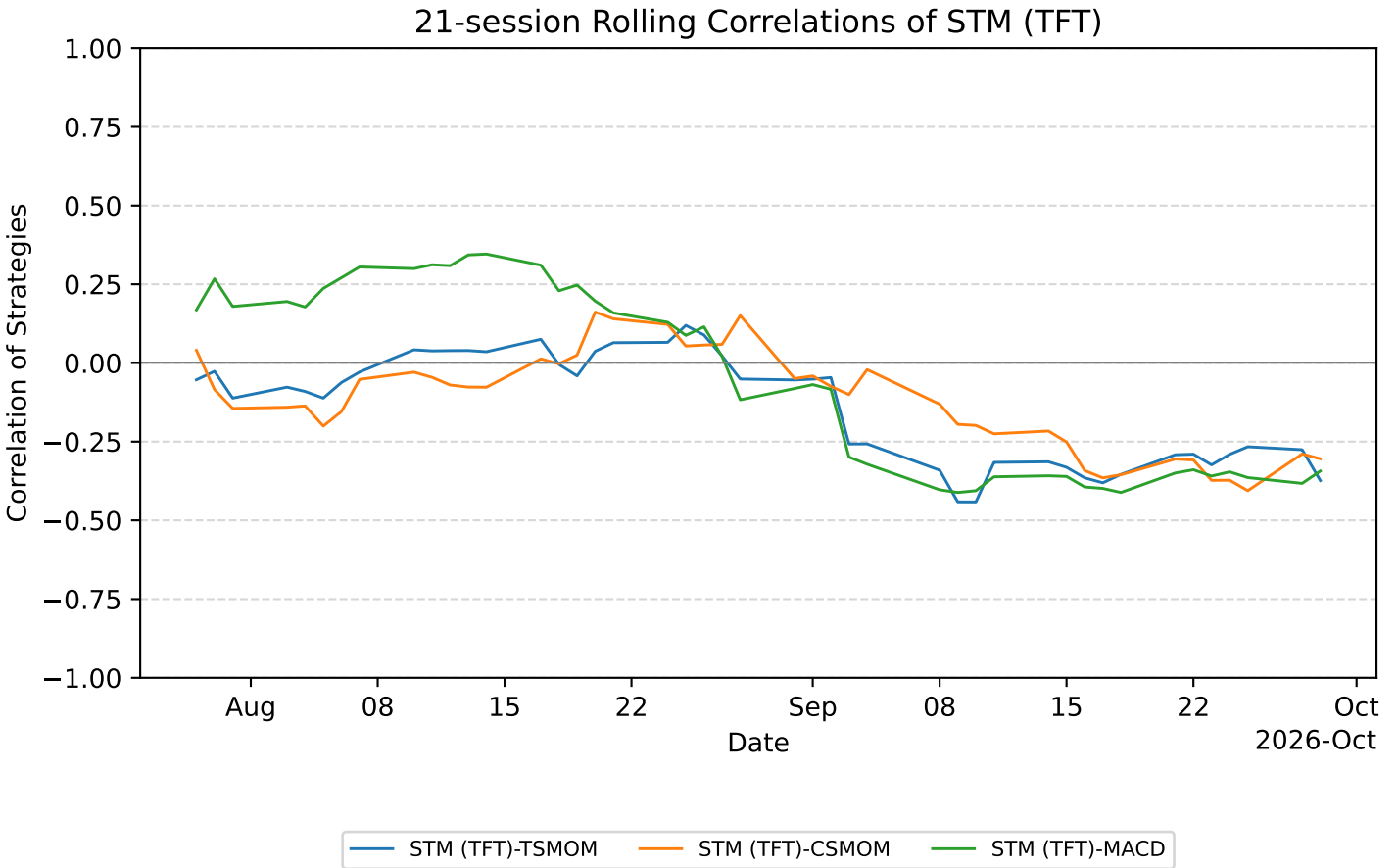
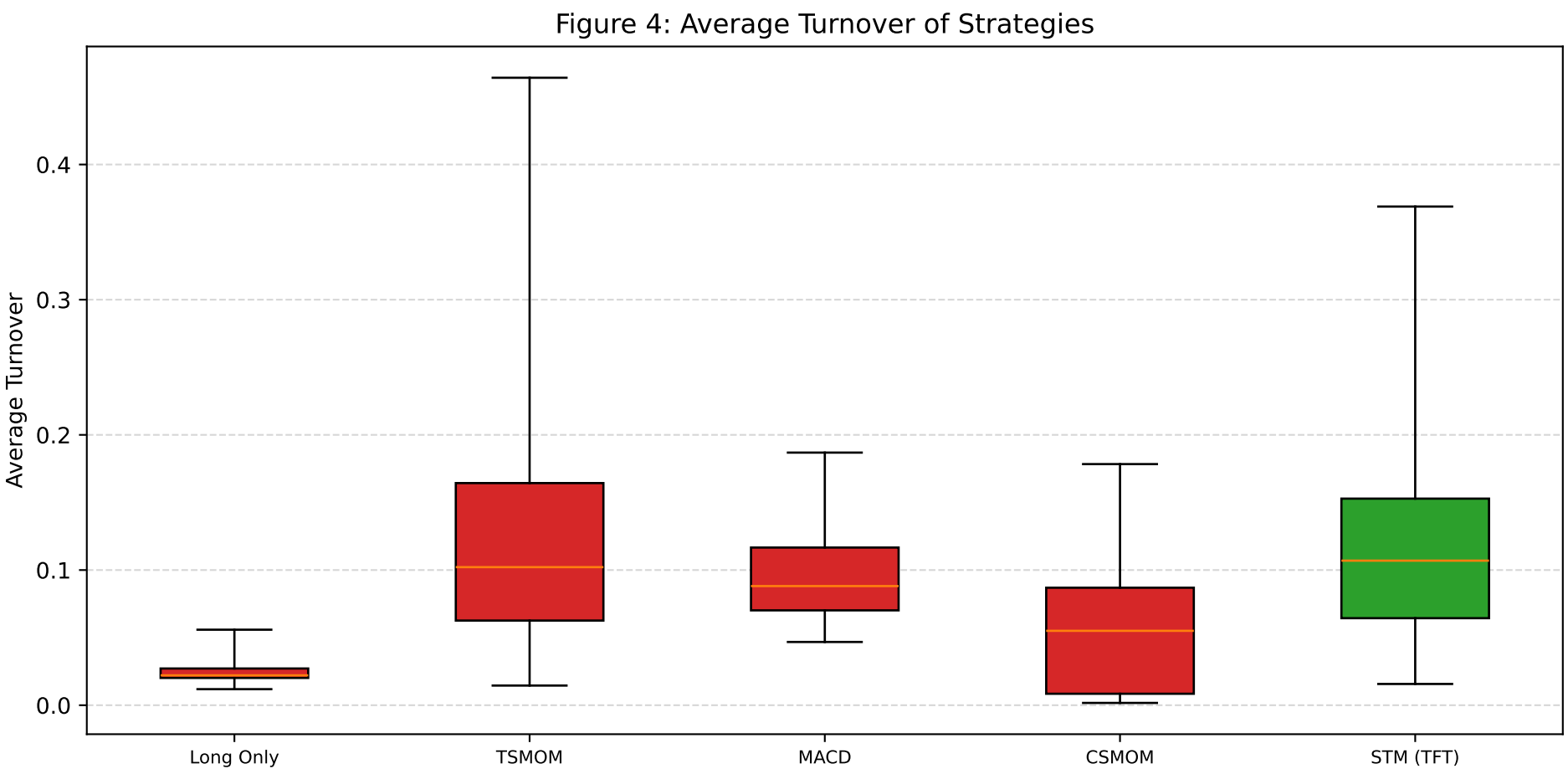
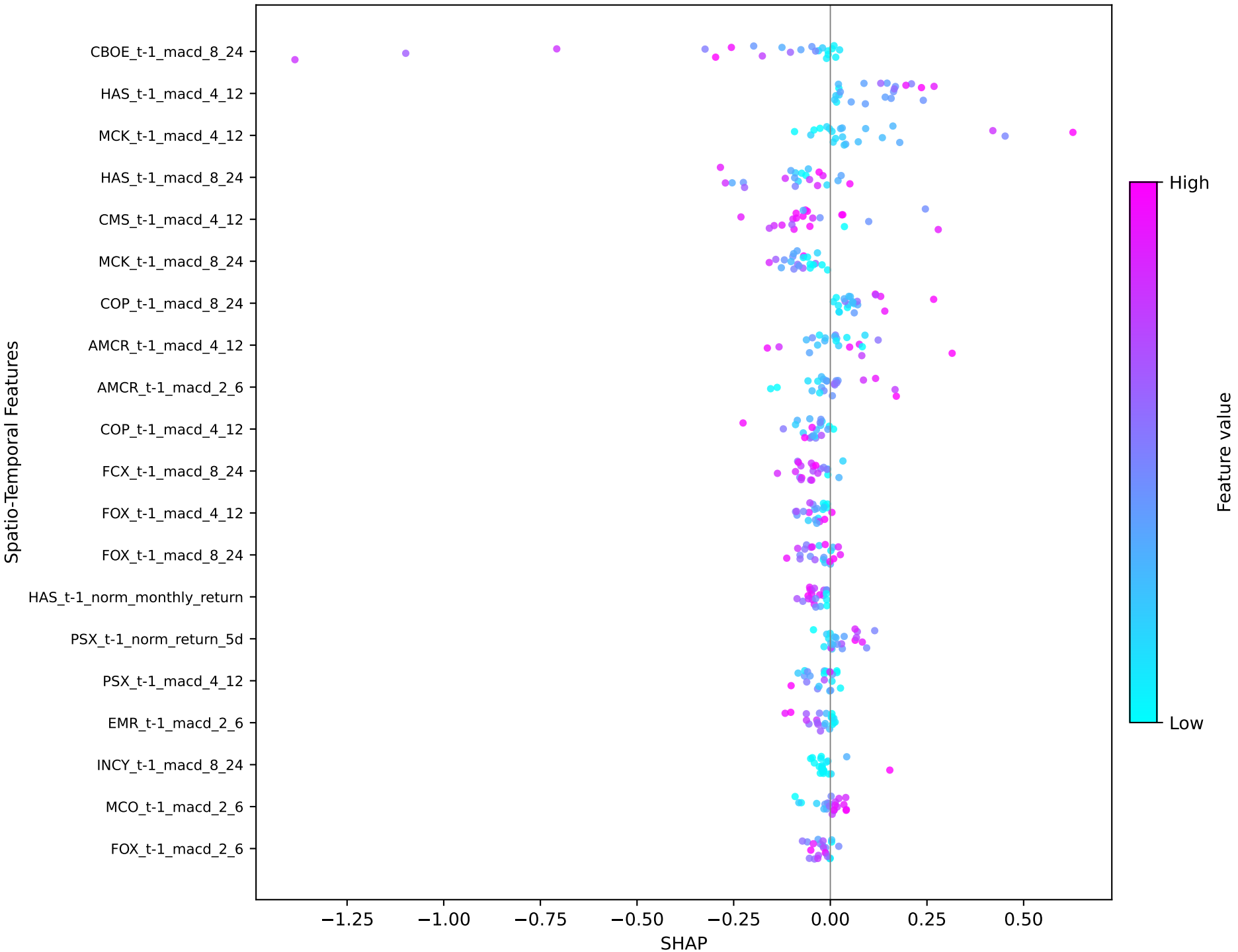
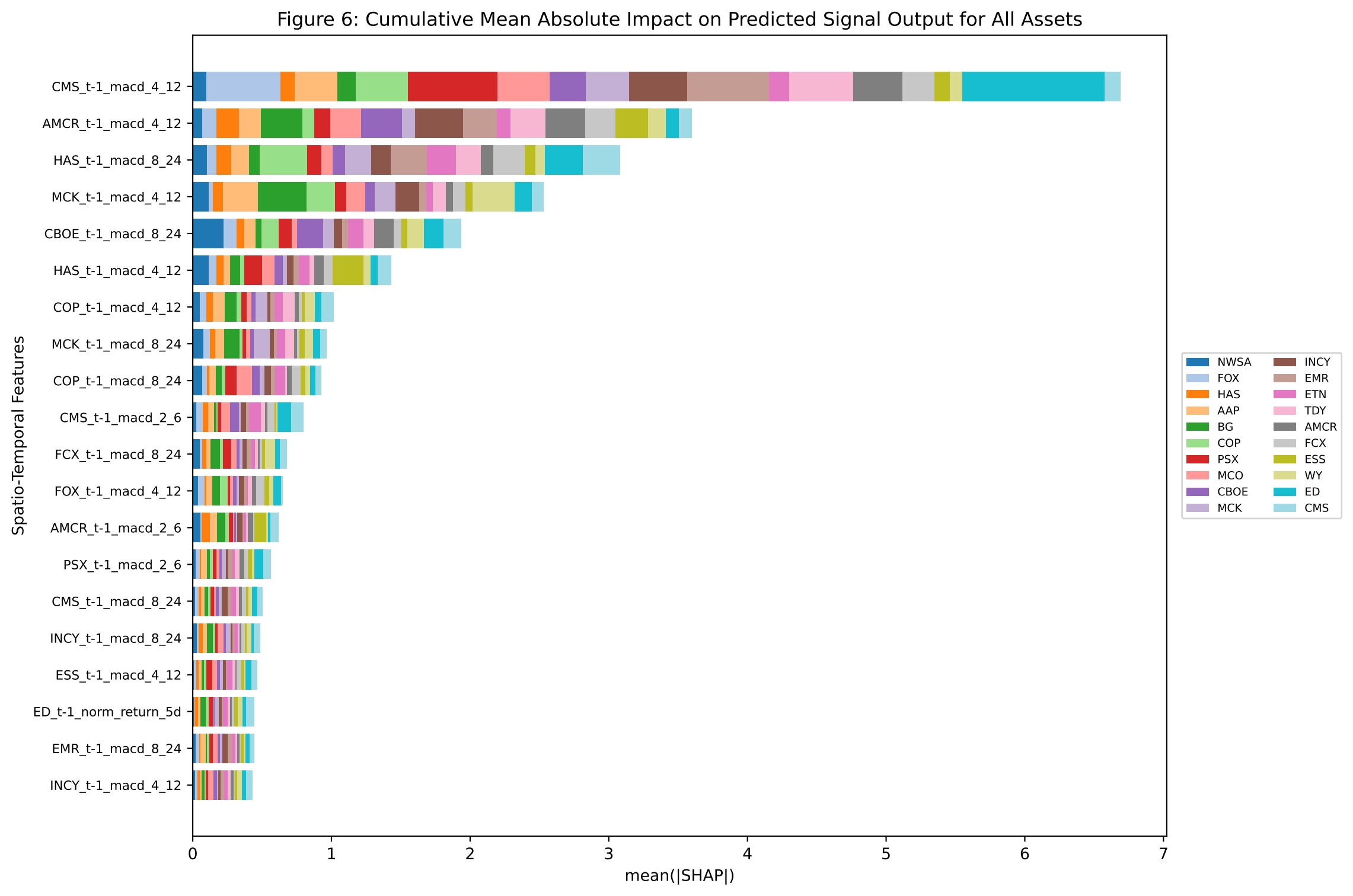


Table 6: Impact of Transactions Costs on Sharpe Ratio – Rescaled to Target Volatility (columns: transaction cost in basis points)

	0	0.5	1	2	3	4	5	10
Long Only	-0.464	-0.470	-0.476	-0.488	-0.500	-0.513	-0.525	-0.585
TSMOM	0.582	0.554	0.527	0.472	0.417	0.363	0.308	0.035
MACD	0.264	0.200	0.136	0.007	-0.123	-0.252	-0.381	-1.029
CSMOM	-1.246	-1.287	-1.329	-1.412	-1.494	-1.577	-1.660	-2.072
STM (TFT)	0.558	0.475	0.393	0.229	0.066	-0.097	-0.259	-1.056


Figure 4: Average Turnover of Strategies
Average Turnover
0.0
0.1
0.2
0.3
0.4
Long Only
TSMOM
MACD
CSMOM
STM (TFT)


Figure 5: Impact on Predicted Signal Output for a Single Asset – NWSA
Spatio-Temporal Features
CBOE_t-1_macd_8_24
HAS_t-1_macd_4_12
MCK_t-1_macd_4_12
HAS_t-1_macd_8_24
CMS_t-1_macd_4_12
MCK_t-1_macd_8_24
COP_t-1_macd_8_24
AMCR_t-1_macd_4_12
AMCR_t-1_macd_2_6
COP_t-1_macd_4_12
FCX_t-1_macd_8_24
FOX_t-1_macd_4_12
FOX_t-1_macd_8_24
HAS_t-1_norm_monthly_return
PSX_t-1_norm_return_5d
PSX_t-1_macd_4_12
EMR_t-1_macd_2_6
INCY_t-1_macd_8_24
MCO_t-1_macd_2_6
FOX_t-1_macd_2_6
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
SHAP
High
Low
Feature value


Figure 6: Cumulative Mean Absolute Impact on Predicted Signal Output for All Assets
Spatio-Temporal Features
CMS_t-1_macd_4_12
AMCR_t-1_macd_4_12
HAS_t-1_macd_8_24
MCK_t-1_macd_4_12
CBOE_t-1_macd_8_24
HAS_t-1_macd_4_12
COP_t-1_macd_4_12
MCK_t-1_macd_8_24
COP_t-1_macd_8_24
CMS_t-1_macd_2_6
FCX_t-1_macd_8_24
FOX_t-1_macd_4_12
AMCR_t-1_macd_2_6
PSX_t-1_macd_2_6
CMS_t-1_macd_8_24
INCY_t-1_macd_8_24
ESS_t-1_macd_4_12
ED_t-1_norm_return_5d
EMR_t-1_macd_8_24
INCY_t-1_macd_4_12
0
1
2
3
4
5
6
7
mean(|SHAP|)
NWSA
FOX
HAS
AAP
BG
COP
PSX
MCO
CBOE
MCK
INCY
EMR
ETN
TDY
AMCR
FCX
ESS
WY
ED
CMS